Bootstrap Framework

Bootstrap Framework

1.1 What is Bootstrap?

Bootstrap is a free, open-source *front-end framework* used to design responsive, mobile-first websites and web applications quickly. Think of it as a giant toolbox filled with pre-built components, styles, and JavaScript plugins. Instead of writing hundreds of lines of CSS from scratch for buttons, navbars, grids, and modals, you simply attach Bootstrap's ready-made **classes** to your HTML elements.

For example, writing `<button class="btn btn-primary">Click</button>` instantly gives you a polished, responsive button with hover effects. Writing `<div class="container">` gives you a centered, properly padded wrapper that adapts to screen sizes. Bootstrap handles all the tedious CSS for you.

1.2 A Brief History

- Created by **Mark Otto** and **Jacob Thornton** at Twitter in **2011**.
- Originally named "**Twitter Blueprint**", later renamed **Bootstrap**.
- **Bootstrap 2** — added the responsive grid and glyphicons.
- **Bootstrap 3** — introduced the mobile-first approach and flat design.
- **Bootstrap 4** — moved to Flexbox, added Sass variables and new utility classes.
- **Bootstrap 5** — removed jQuery (pure vanilla JavaScript), added CSS custom properties, RTL support, and a new utilities API. This is the current major version.

1.3 Key Features of Bootstrap

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- **Mobile-first:** base styles are written for small screens, then scale up for larger devices.
- **Responsive grid system:** a flexible 12-column flexbox grid that adapts to any screen.
- **Prebuilt components:** navbar, cards, modals, alerts, accordions, dropdowns, and many more.
- **Utility classes:** small single-purpose classes for spacing, colors, display, text alignment, etc.
- **JavaScript plugins:** dropdowns, tabs, tooltips, modals work without writing any JavaScript.
- **Customizable:** can be modified using Sass variables (colors, spacing, breakpoints).
- **Cross-browser compatible:** works on all modern browsers (Chrome, Firefox, Safari, Edge).
- **Huge community:** extensive documentation, tutorials, and third-party themes.

1.4 Advantages and Disadvantages

Advantages	Disadvantages
Very fast development — saves a lot of time.	Websites can start looking similar to each other (Bootstrap "look").
Responsive design comes built-in, no extra effort.	Large file size if you include the full CSS and JS.
Consistent look and feel across all pages.	Requires learning many class names (but they are logical).
Huge community and excellent documentation.	Heavy customization may need Sass knowledge.
Beginner-friendly — no advanced CSS required.	Too many classes can make HTML look cluttered.

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Accessible components out of the box.	Some JavaScript plugins require Popper.js (included in the bundle).

1.5 How to Include Bootstrap in a Project

There are three common ways to add Bootstrap to your project:

Method 1 — CDN (Content Delivery Network) — easiest

Just add the following lines to your HTML. This is the fastest way to get started.

```
<!-- CSS in the <head> -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstr
      rel="stylesheet">

<!-- JavaScript bundle just before </body> -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstr
```

Method 2 — Download the files locally

Download the compiled ZIP from the official website, then link `bootstrap.min.css` and `bootstrap.bundle.min.js` from your local `css/` and `js/` folders.

```
<link rel="stylesheet" href="css/bootstrap.min.css">
<script src="js/bootstrap.bundle.min.js"></script>
```

Method 3 — Using npm / package managers

For modern build tools (Webpack, Vite, etc.):

```
npm install bootstrap
# Then import in your JS:
# import 'bootstrap/dist/css/bootstrap.min.css';
# import 'bootstrap/dist/js/bootstrap.bundle.min.js';
```

> **Note:** The `bootstrap.bundle.min.js` includes Popper.js, which is required for dropdowns, tooltips, and popovers. If you use the separate `bootstrap.min.js`, you

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must also include Popper.js separately.

1.6 Basic Bootstrap Page Template

Every Bootstrap page should start with this structure:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <!-- REQUIRED for responsive behaviour -->
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>My Bootstrap Page</title>
  <link href="bootstrap.min.css" rel="stylesheet">
</head>
<body>

  <div class="container">
    <h1>Hello Bootstrap!</h1>
    <button class="btn btn-primary">Click Me</button>
  </div>

  <script src="bootstrap.bundle.min.js"></script>
</body>
</html>
```

Important: The `<meta name="viewport">` tag is *mandatory* for responsive behaviour. Without it, media queries will not work properly on mobile devices, and your site will look like a zoomed-out desktop page.

1.7 Bootstrap File Structure (Downloaded Version)

```
bootstrap/
├── css/
│   ├── bootstrap.css
│   ├── bootstrap.min.css
│   ├── bootstrap.rtl.min.css   (for right-to-left languages)
│   └── bootstrap-grid.min.css  (only the grid, smaller)
├── js/
│   ├── bootstrap.bundle.js
```

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```
│   ├── bootstrap.bundle.min.js
│   ├── bootstrap.js
│   └── bootstrap.min.js
└── icons/  (Bootstrap Icons font files)
```

1.8 How Bootstrap Works Under the Hood

- **CSS classes:** Bootstrap provides hundreds of classes. When you add them to HTML, the corresponding CSS rules apply.
- **Flexbox & Grid:** The layout system is built on CSS Flexbox. The grid uses flex rows and columns.
- **JavaScript plugins:** Written in vanilla JavaScript (v5). They use HTML5 data attributes like `data-bs-toggle` to attach behaviour without extra code.
- **CSS custom properties:** Bootstrap 5 uses CSS variables (e.g., `--bs-primary`) for easy theming.
- **Sass:** The source files are written in Sass, allowing deep customization before compiling.

1.9 Bootstrap Icons (Bonus)

Bootstrap also has a free icon library. Include it via CDN:

```
<link rel="stylesheet"
      href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.3/font/boot

<!-- Usage -->
<i class="bi bi-alarm"></i>
<i class="bi bi-heart-fill text-danger"></i>
```

2. Bootstrap Layout

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Layout is the backbone of Bootstrap. It decides **where** things appear on the page. It is built on three things: **containers**, **rows**, and **columns**, powered by **Flexbox**. Once you understand these three, you can build almost any page structure.

2.1 Containers

A container is a wrapper that holds your content and gives it a proper width, padding, and centers it on the screen. Bootstrap provides three main container types:

Class	Behaviour	When to use
`.container`	Fixed max-width at each breakpoint; horizontally centered.	Most websites (default choice).
`.container-fluid`	Always 100% width of the viewport (full width).	Full-width sections, dashboards, landing pages.
`.container-{breakpoint}` (e.g. `.container-md`)	100% wide until that breakpoint, then fixed width.	Mixed layouts where you want full width on small screens.

```
<div class="container"> ... </div>
<div class="container-fluid"> ... </div>
<div class="container-md"> ... </div>
```

> **Note:** Containers have built-in horizontal padding (called "gutters") of 12px on each side, so content doesn't touch the screen edges. Rows add negative margins to offset this padding.

2.2 The 12-Column Grid System

Bootstrap divides the page into **12 equal columns**. Why 12? Because 12 is divisible by 2, 3, 4, and 6, making it easy to create many layouts (halves, thirds, quarters, sixths). You wrap columns inside a `.row`, and the row must be inside a container.

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- `.row` — a horizontal group of columns (a flex container).
- `.col` — auto-width column (all columns share space equally).
- `.col-4` — a column taking 4 of the 12 parts (i.e. 33.33% width).

```
<div class="container">
  <div class="row">
    <div class="col-4">Column 1 (4/12)</div>
    <div class="col-4">Column 2 (4/12)</div>
    <div class="col-4">Column 3 (4/12)</div>
  </div>
</div>
```

The 12 numbers always add up to 12. Some valid splits:

- `col-12` → full width
- `col-6 col-6` → two halves
- `col-4 col-4 col-4` → three equal parts
- `col-3 col-3 col-3 col-3` → four equal parts
- `col-8 col-4` → main content + sidebar
- `col-2 col-8 col-2` → narrow left, wide center, narrow right

2.3 Breakpoints and Responsive Classes

Bootstrap uses six breakpoints. Responsive classes are written as `.col-{breakpoint}-{size}` and apply **from that breakpoint and above** (mobile-first). This means you only need to set the smallest breakpoint and then override as the screen grows.

Breakpoint	Class infix	Viewport width	Typical device
Extra small	*(none)*	< 576px	Phones
Small	`sm`	≥ 576px	Large phones
Medium	`md`	≥ 768px	Tablets
Large	`lg`	≥ 992px	Laptops
Extra large	`xl`	≥ 1200px	Desktops

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Extra extra large	`xxl`	≥ 1400px	Large screens

```
<div class="row">
  <!-- 12 columns on mobile, 6 on tablet, 3 on desktop -->
  <div class="col-12 col-md-6 col-lg-3">Box 1</div>
  <div class="col-12 col-md-6 col-lg-3">Box 2</div>
  <div class="col-12 col-md-6 col-lg-3">Box 3</div>
  <div class="col-12 col-md-6 col-lg-3">Box 4</div>
</div>
```

> **Tip:** Mobile-first means the *smallest* class is the base and larger breakpoints override it. So `col-12 col-md-6` means "full width on phones, half width from tablets up".

2.4 Auto-layout Columns

Use `.col` (without a number) to let columns share the remaining space equally.

```
<div class="row">
  <div class="col">Equal</div>
  <div class="col">Equal</div>
  <div class="col">Equal</div>
</div>

<!-- One column with fixed width, others share the rest -->
<div class="row">
  <div class="col-4">Fixed 4/12</div>
  <div class="col">Auto</div>
  <div class="col">Auto</div>
</div>
```

2.5 Row Columns (setting columns per row)

Instead of adding a width class to every column, you can use `.row-cols-*` on the row to set how many columns appear per row. This is very handy for card galleries.

```
<div class="row row-cols-1 row-cols-md-3 g-4">
  <div class="col">Card 1</div>
  <div class="col">Card 2</div>
```

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```
  <div class="col">Card 3</div>
  <div class="col">Card 4</div>
</div>
```

2.6 Gutters (space between columns)

Gutters are the gaps between columns. Use `g-*` (all), `gx-*` (horizontal only), or `gy-*` (vertical only) on the row. Values range from `g-0` to `g-5`.

```
<div class="row g-4">
  <div class="col-6">...</div>
  <div class="col-6">...</div>
</div>

<!-- No horizontal gap, only vertical -->
<div class="row gx-0 gy-3"> ... </div>
```

2.7 Nesting

You can place a `.row` inside a `.col` to create sub-grids.

```
<div class="row">
  <div class="col-6">
    <div class="row">
      <div class="col-6">Nested A</div>
      <div class="col-6">Nested B</div>
    </div>
  </div>
  <div class="col-6">Outer right</div>
</div>
```

2.8 Alignment (Flexbox utilities in the grid)

Since rows are flex containers, you can use flex utility classes to align columns vertically and horizontally.

- **Vertical alignment:** `align-items-start`, `align-items-center`, `align-items-end`, `align-items-stretch` on the row.

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- **Individual column vertical alignment:** `align-self-start`, `align-self-center`, `align-self-end`.
- **Horizontal distribution:** `justify-content-start`, `justify-content-center`, `justify-content-end`, `justify-content-between`, `justify-content-around`, `justify-content-evenly`.

```
<div class="row align-items-center" style="height: 150px;">
  <div class="col">Vertically centered</div>
  <div class="col">Vertically centered</div>
</div>

<div class="row justify-content-center">
  <div class="col-4">Centered horizontally</div>
</div>
```

2.9 Offset Columns

Offsets push a column to the right by leaving empty grid columns before it. The class format is `.offset-{number}` or `.offset-{breakpoint}-{number}`.

```
<div class="row">
  <div class="col-md-4">First</div>
  <div class="col-md-4 offset-md-4">Pushed to the right</div>
</div>

<!-- Centered box: 6 columns + 3 offset = 9, remaining 3 on right -->
<div class="row">
  <div class="col-md-6 offset-md-3">Centered content</div>
</div>
```

Class	Meaning
`offset-3`	Shift right by 3 columns on all screens.
`offset-md-3`	Shift right by 3 columns from tablet upward.
`offset-md-0`	Remove the offset from tablet upward.

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2.10 Reordering Columns

Reordering changes the visual order of columns without changing the HTML order. This is very useful for responsive designs (e.g., moving a sidebar below the content on mobile).

Class	Meaning
`order-0` … `order-5`	Set order priority (lower = first).
`order-first`	Order: -1 (always first).
`order-last`	Order: 6 (always last).
`order-md-2`	Order 2 from tablet upward.

```
<div class="row">
  <div class="col-md-8 order-md-2">Main Content (shown 2nd on desktop)</d
  <div class="col-md-4 order-md-1">Sidebar (shown 1st on desktop)</div>
</div>
```

> **Offset vs Order:** `offset-*` adds *empty space* before a column; `order-*` *changes the sequence* of columns without adding space.

2.11 Flex Utilities in Layout (d-flex)

Bootstrap's grid is flex-based, but you can also use flex utilities directly on any container:

- `d-flex` / `d-inline-flex` — make an element a flex container.
- `flex-row` / `flex-column` — direction.
- `flex-row-reverse` / `flex-column-reverse` — reverse direction.
- `flex-wrap` / `flex-nowrap` — wrapping.
- `justify-content-*` — horizontal distribution.
- `align-items-*` — vertical alignment.
- `flex-grow-1` — let an item grow.
- `gap-*` — space between flex items.

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```
<div class="d-flex justify-content-between align-items-center gap-3">
  <div>Left</div>
  <div>Right</div>
</div>
```

3. Bootstrap Content

"Content" means the default styling Bootstrap applies to normal HTML elements — headings, paragraphs, tables, images, and form controls. Bootstrap styles these elements so they look clean and consistent without extra CSS.

3.1 Typography

Bootstrap automatically styles HTML headings, paragraphs, lists, and blockquotes. You can also use utility classes to modify text.

Headings

```
<h1>Heading 1</h1>   <!-- 2.5rem -->
<h2>Heading 2</h2>   <!-- 2rem -->
<h3>Heading 3</h3>   <!-- 1.75rem -->
<h4>Heading 4</h4>   <!-- 1.5rem -->
<h5>Heading 5</h5>   <!-- 1.25rem -->
<h6>Heading 6</h6>   <!-- 1rem -->

<!-- Make a span look like a heading -->
<p class="h3">Looks like an h3</p>
```

Display Headings

Larger, bolder headings used for hero sections:

```
<h1 class="display-1">Display 1</h1>
<h1 class="display-2">Display 2</h1>
<h1 class="display-3">Display 3</h1>
<h1 class="display-4">Display 4</h1>
```

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```
<h1 class="display-5">Display 5</h1>
<h1 class="display-6">Display 6</h1>
```

Lead paragraph

```
<p class="lead">This paragraph stands out slightly larger.</p>
```

Inline text elements

```
<p>You can use <mark>highlight</mark>, <del>deleted</del>,
<s>strikethrough</s>, <ins>inserted</ins>, <u>underlined</u>,
<small>small</small>, <strong>bold</strong>, <em>italic</em>.</p>
```

Text emphasis classes

Class	Effect
`.text-muted`	Grey, secondary text
`.fw-bold` / `.fw-semibold` / `.fw-normal` / `.fw-light`	Font weight
`.fst-italic` / `.fst-normal`	Font style
`.text-decoration-underline` / `-line-through` / `-none`	Decoration
`.text-lowercase` / `.text-uppercase` / `.text-capitalize`	Case change
`.text-start` / `.text-center` / `.text-end`	Alignment
`.text-wrap` / `.text-nowrap`	Wrapping behaviour
`.text-break`	Breaks long words
`.text-truncate`	Adds "…" when text overflows
`.fs-1` … `.fs-6`	Font size
`.lh-1` / `.lh-sm` / `.lh-lg`	Line height

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`.font-monospace`	Monospace font

Blockquotes

```
<blockquote class="blockquote">
  <p>The best way to predict the future is to create it.</p>
  <footer class="blockquote-footer">Peter Drucker</footer>
</blockquote>
```

Lists

```
<ul class="list-unstyled">   <!-- removes bullets and left padding -->
  <li>Item one</li>
</ul>

<ul class="list-inline">     <!-- horizontal list -->
  <li class="list-inline-item">Home</li>
  <li class="list-inline-item">About</li>
</ul>
```

Code and Keyboard input

```
<p>Use <code>const</code> to declare a constant.</p>
<p>Press <kbd>Ctrl</kbd> + <kbd>C</kbd> to copy.</p>
<pre><code>console.log('Hello');</code></pre>
```

3.2 Tables

Add the base class `.table` to any `<table>` to get Bootstrap styling.

Class	Description
`.table`	Base table styling (padding, borders, alignment)
`.table-striped`	Zebra-striped rows (alternate background)
`.table-bordered`	Borders on all sides of every cell

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`.table-borderless`	No borders at all
`.table-hover`	Row highlight on mouse hover
`.table-sm`	Compact table with less padding
`.table-dark`	Dark background table
`.table-responsive`	Adds horizontal scrolling on small screens
`.table-group-divider`	Thicker divider between thead/tbody
`.table-primary` … `.table-danger`	Colored rows/cells
`.table caption-top`	Caption above the table

```
<div class="table-responsive">
  <table class="table table-striped table-hover table-bordered">
    <caption>List of employees</caption>
    <thead class="table-dark">
      <tr>
        <th>#</th>
        <th>Name</th>
        <th>City</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>1</td>
        <td>Amit</td>
        <td>Delhi</td>
      </tr>
      <tr class="table-success">
        <td>2</td>
        <td>Priya</td>
        <td>Mumbai</td>
      </tr>
    </tbody>
    <tfoot>
      <tr><td colspan="3">Total: 2 employees</td></tr>
    </tfoot>
```

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```
  </table>
</div>
```

> **Tip:** Always wrap tables inside `.table-responsive` so they scroll horizontally on mobile instead of breaking the layout. You can also use `.table-responsive-sm`, `.table-responsive-md`, etc. to target specific breakpoints.

3.3 Images

Class	Effect
`.img-fluid`	Makes the image responsive: `max-width:100%; height:auto;`
`.img-thumbnail`	Adds a rounded border and padding (thumbnail look)
`.rounded` / `.rounded-circle` / `.rounded-pill`	Rounded corners
`.float-start` / `.float-end`	Float left / right
`.mx-auto d-block`	Center an image

```
<img src="photo.jpg" class="img-fluid rounded" alt="Sample">

<!-- Thumbnail -->
<img src="photo.jpg" class="img-thumbnail" alt="Thumb">

<!-- Circular avatar -->
<img src="user.jpg" class="rounded-circle" width="80" height="80" alt="User">

<!-- Figure with caption -->
<figure class="figure">
  <img src="chart.png" class="figure-img img-fluid rounded" alt="Chart">
  <figcaption class="figure-caption">Monthly sales report.</figcaption>
</figure>
```

3.4 Forms

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Bootstrap forms are clean and stacked by default. Main classes:

Class	Purpose
`.form-label`	Styles a `<label>`
`.form-control`	Styles text inputs, textareas, file inputs
`.form-select`	Styles `<select>` dropdowns
`.form-check`	Wrapper for checkboxes and radios
`.form-check-input`	The checkbox / radio itself
`.form-check-label`	Its label
`.form-switch`	Turns a checkbox into a toggle switch
`.input-group`	Combines input with buttons or text
`.form-text`	Small helper text below an input
`.form-control-sm` / `.form-control-lg`	Small / large inputs
`.is-valid` / `.is-invalid`	Green / red validation state
`.form-floating`	Floating label style
`.form-range`	Styles a range slider

Basic form

```
<form>
  <div class="mb-3">
    <label for="email" class="form-label">Email address</label>
    <input type="email" class="form-control" id="email"
           placeholder="name@example.com">
    <div class="form-text">We never share your email.</div>
  </div>

  <div class="mb-3">
    <label for="city" class="form-label">City</label>
```

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```
    <select class="form-select" id="city">
      <option>Delhi</option>
      <option>Mumbai</option>
    </select>
  </div>

  <div class="form-check mb-3">
    <input class="form-check-input" type="checkbox" id="agree">
    <label class="form-check-label" for="agree">I agree</label>
  </div>

  <button type="submit" class="btn btn-primary">Submit</button>
</form>
```

Floating labels

```
<div class="form-floating mb-3">
  <input type="email" class="form-control" id="emailFloat"
         placeholder="name@example.com">
  <label for="emailFloat">Email address</label>
</div>
```

Input group

```
<div class="input-group mb-3">
  <span class="input-group-text">@</span>
  <input type="text" class="form-control" placeholder="username">
</div>
```

Validation

```
<input class="form-control is-invalid" type="text">
<div class="invalid-feedback">Please enter a valid value.</div>

<input class="form-control is-valid" type="text">
<div class="valid-feedback">Looks good!</div>
```

Switch

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```
<div class="form-check form-switch">
  <input class="form-check-input" type="checkbox" id="sw">
  <label class="form-check-label" for="sw">Enable notifications</label>
</div>
```

Horizontal form layout (using the grid)

```
<form class="row g-3">
  <div class="col-md-4">
    <label for="fname" class="form-label">First name</label>
    <input type="text" class="form-control" id="fname">
  </div>
  <div class="col-md-4">
    <label for="lname" class="form-label">Last name</label>
    <input type="text" class="form-control" id="lname">
  </div>
  <div class="col-12">
    <button class="btn btn-primary" type="submit">Submit</button>
  </div>
</form>
```

4. Bootstrap Components

Components are ready-made UI blocks (navbar, cards, modals, etc.). Most of them work with just HTML classes, and some need Bootstrap's JavaScript (included in the bundle file).

4.1 Navbar

The navbar is the top navigation bar. It is responsive: on small screens the links collapse into a hamburger button.

Class	Purpose
`.navbar`	Base navbar container
`.navbar-expand-lg`	Expands horizontally from the `lg` breakpoint up

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`.navbar-brand`	Logo / site name
`.navbar-nav`	Container for nav links
`.nav-item` / `.nav-link`	Individual link item
`.navbar-toggler`	Hamburger button (mobile)
`.collapse .navbar-collapse`	Collapsible menu wrapper
`.active`	Highlights the current page link
`.navbar-dark`	Light text on dark backgrounds

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">

    <a class="navbar-brand" href="#">MySite</a>

    <button class="navbar-toggler" type="button"
            data-bs-toggle="collapse" data-bs-target="#mainNav">
      <span class="navbar-toggler-icon"></span>
    </button>

    <div class="collapse navbar-collapse" id="mainNav">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item">
          <a class="nav-link active" href="#">Home</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="#">About</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="#">Contact</a>
        </li>
      </ul>
    </div>

  </div>
</nav>
```

Useful navbar helpers:

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- `.fixed-top` — sticks the navbar to the top permanently.
- `.sticky-top` — sticks after scrolling past it.
- `.bg-primary`, `.bg-dark`, `.bg-light` — background colors.
- `.navbar-dark` — light text (for dark backgrounds).
- `.ms-auto` — pushes the nav links to the right.
- `.me-auto` — pushes the nav links to the left.

4.2 Navs and Tabs

The `.nav` component creates simple navigation lists. It can be shown as **tabs** or **pills**, and can even switch content panels.

Class	Purpose
`.nav`	Base nav container
`.nav-item`	Each list item
`.nav-link`	Each clickable link
`.nav-tabs`	Tab-style appearance
`.nav-pills`	Rounded button-style appearance
`.active`	Highlights the selected item
`.disabled`	Greys out and disables an item
`.nav-fill` / `.nav-justified`	Stretch items to fill width
`.flex-column`	Vertical navigation
`.nav-underline`	Underline style (v5.3+)

```
<!-- Simple pills -->
<ul class="nav nav-pills">
  <li class="nav-item">
    <a class="nav-link active" href="#">Home</a>
```

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```
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Profile</a>
  </li>
  <li class="nav-item">
    <a class="nav-link disabled" href="#">Disabled</a>
  </li>
</ul>
```

Tabs that switch content

```
<ul class="nav nav-tabs" id="myTab">
  <li class="nav-item">
    <button class="nav-link active" data-bs-toggle="tab"
            data-bs-target="#tab1">Tab 1</button>
  </li>
  <li class="nav-item">
    <button class="nav-link" data-bs-toggle="tab"
            data-bs-target="#tab2">Tab 2</button>
  </li>
</ul>

<div class="tab-content">
  <div class="tab-pane fade show active" id="tab1">Content of tab 1</div>
  <div class="tab-pane fade" id="tab2">Content of tab 2</div>
</div>
```

4.3 Dropdowns

A dropdown is a toggleable menu. It requires the Bootstrap JS bundle.

Class / Attribute	Purpose
`.dropdown`	Wrapper element
`.dropdown-toggle`	The button that opens the menu
`.dropdown-menu`	The menu container
`.dropdown-item`	Each menu link

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`.dropdown-divider`	Horizontal separator line
`.dropdown-header`	Non-clickable heading
`.dropdown-item-text`	Plain text inside the menu
`.dropup` / `.dropend` / `.dropstart`	Menu direction
`.dropdown-menu-end`	Right-align the menu
`data-bs-toggle="dropdown"`	Activates the dropdown

```
<div class="dropdown">
  <button class="btn btn-secondary dropdown-toggle" type="button"
          data-bs-toggle="dropdown">
    Choose Option
  </button>

  <ul class="dropdown-menu">
    <li><h6 class="dropdown-header">Actions</h6></li>
    <li><a class="dropdown-item" href="#">Edit</a></li>
    <li><a class="dropdown-item" href="#">Duplicate</a></li>
    <li><hr class="dropdown-divider"></li>
    <li><a class="dropdown-item text-danger" href="#">Delete</a></li>
  </ul>
</div>
```

4.4 Buttons

Base class `.btn` + a variant class. Bootstrap 5 provides 8 solid and 8 outline variants.

Class	Meaning
`.btn-primary`	Main action (blue)
`.btn-secondary`	Secondary action (grey)
`.btn-success`	Positive action (green)
`.btn-danger`	Delete / destructive (red)

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`.btn-warning`	Caution (yellow)
`.btn-info`	Informational (cyan)
`.btn-light` / `.btn-dark`	Light / dark
`.btn-link`	Looks like a hyperlink
`.btn-outline-primary` …	Transparent with colored border
`.btn-lg` / `.btn-sm`	Large / small size
`.disabled` or `disabled` attribute	Disabled state
`.w-100` or `.d-grid`	Full-width (block) button

```
<button class="btn btn-primary">Primary</button>
<button class="btn btn-danger">Danger</button>
<button class="btn btn-outline-success">Outline Success</button>
<button class="btn btn-warning btn-lg">Large Warning</button>
<a href="#" class="btn btn-dark">Link as Button</a>
<button class="btn btn-primary" disabled>Disabled</button>
```

4.5 Button Groups

Groups several buttons together into one connected bar.

Class	Purpose
`.btn-group`	Horizontal group
`.btn-group-vertical`	Vertical group
`.btn-group-lg` / `.btn-group-sm`	Size of the whole group
`.btn-toolbar`	Groups multiple button groups together

```
<div class="btn-group" role="group">
  <button class="btn btn-outline-primary">Left</button>
```

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```
  <button class="btn btn-outline-primary">Middle</button>
  <button class="btn btn-outline-primary">Right</button>
</div>

<div class="btn-group-vertical">
  <button class="btn btn-secondary">Top</button>
  <button class="btn btn-secondary">Bottom</button>
</div>
```

4.6 Breadcrumb

Shows the user's position in the site hierarchy (Home / Products / Shoes).

```
<nav aria-label="breadcrumb">
  <ol class="breadcrumb">
    <li class="breadcrumb-item"><a href="#">Home</a></li>
    <li class="breadcrumb-item"><a href="#">Products</a></li>
    <li class="breadcrumb-item active" aria-current="page">Shoes</li>
  </ol>
</nav>
```

The separator is created by CSS. The last item should have `.active` and `aria-current="page"`.

4.7 Pagination

Used to split long lists of content into pages.

Class	Purpose
`.pagination`	Wrapper (a `<ul>`)
`.page-item`	Each `<li>`
`.page-link`	Each clickable `<a>`
`.active`	Current page
`.disabled`	Disabled (e.g. "Previous" on page 1)

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`.pagination-lg` / `.pagination-sm`	Sizes
`.justify-content-center`	Center the pagination

```
<nav>
  <ul class="pagination justify-content-center">
    <li class="page-item disabled">
      <a class="page-link" href="#">Previous</a>
    </li>
    <li class="page-item active">
      <a class="page-link" href="#">1</a>
    </li>
    <li class="page-item">
      <a class="page-link" href="#">2</a>
    </li>
    <li class="page-item">
      <a class="page-link" href="#">Next</a>
    </li>
  </ul>
</nav>
```

4.8 Labels (Badges)

In Bootstrap 5, "labels" from Bootstrap 3 are now called **badges** (`.badge`). They are small colored counters or tags.

```
<h4>Inbox <span class="badge bg-danger">4</span></h4>

<button class="btn btn-primary">
  Messages <span class="badge bg-light text-dark">12</span>
</button>

<span class="badge rounded-pill bg-success">New</span>
<span class="badge text-bg-warning">Warning</span>
```

Use `.bg-*` or `.text-bg-*` for colors and `.rounded-pill` for pill shape.

4.9 Alerts

Alerts give feedback messages to the user (success, error, warning, info).

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Class	Meaning
`.alert`	Base class (required)
`.alert-success`	Green success message
`.alert-danger`	Red error message
`.alert-warning`	Yellow warning
`.alert-info`	Blue information
`.alert-primary`, `.alert-secondary`, `.alert-dark`	Other variants
`.alert-heading`	A heading inside the alert
`.alert-link`	A link matching the alert color
`.alert-dismissible` + `.btn-close`	Closable alert

```
<div class="alert alert-success">Your data has been saved!</div>

<div class="alert alert-danger alert-dismissible fade show">
  <strong>Error!</strong> Something went wrong.
  <button type="button" class="btn-close" data-bs-dismiss="alert"></butto
</div>

<div class="alert alert-info">
  <h5 class="alert-heading">Heads up!</h5>
  <p>Read this <a href="#" class="alert-link">important link</a>.</p>
</div>
```

4.10 Progress Bars

Show how much of a task is completed.

Class	Purpose
`.progress`	Outer container

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`.progress-bar`	Inner colored bar
`style="width: 60%"`	Sets the progress amount
`.progress-bar-striped`	Striped pattern
`.progress-bar-animated`	Animated stripes
`.bg-success`, `.bg-warning`…	Bar color

```
<div class="progress">
  <div class="progress-bar" style="width: 25%">25%</div>
</div>

<div class="progress mt-3">
  <div class="progress-bar bg-success progress-bar-striped
              progress-bar-animated" style="width: 70%">
    70%
  </div>
</div>

<!-- Stacked (multiple bars in one) -->
<div class="progress">
  <div class="progress-bar" style="width: 35%">35%</div>
  <div class="progress-bar bg-warning" style="width: 25%">25%</div>
</div>
```

Always add `role="progressbar"`, `aria-valuenow`, `aria-valuemin`, and `aria-valuemax` for accessibility.

4.11 Accordion

An accordion is a set of collapsible panels where only one (or more) panel is open at a time.

Class / Attribute	Purpose
`.accordion`	Outer wrapper
`.accordion-item`	Each panel

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`.accordion-header`	Header area
`.accordion-button`	The clickable toggle button
`.accordion-collapse collapse show`	The collapsible body (`show` = open by default)
`.accordion-body`	The content area
`data-bs-parent="#id"`	Ensures only one panel is open at a time
`.accordion-flush`	Removes the outer border

```
<div class="accordion" id="faq">

  <div class="accordion-item">
    <h2 class="accordion-header">
      <button class="accordion-button" type="button"
              data-bs-toggle="collapse" data-bs-target="#q1">
        What is Bootstrap?
      </button>
    </h2>
    <div id="q1" class="accordion-collapse collapse show"
         data-bs-parent="#faq">
      <div class="accordion-body">
        Bootstrap is a CSS framework for responsive websites.
      </div>
    </div>
  </div>

  <div class="accordion-item">
    <h2 class="accordion-header">
      <button class="accordion-button collapsed" type="button"
              data-bs-toggle="collapse" data-bs-target="#q2">
        Is jQuery required?
      </button>
    </h2>
    <div id="q2" class="accordion-collapse collapse"
         data-bs-parent="#faq">
      <div class="accordion-body">
        No, Bootstrap 5 uses pure vanilla JavaScript.
      </div>
    </div>
  </div>
```

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```
</div>
```

4.12 Card

A card is a flexible, bordered box that holds content — image, title, text, links. It is one of the most used Bootstrap components.

Class	Purpose
`.card`	Outer box
`.card-body`	Padding wrapper for content
`.card-title`	Heading inside card
`.card-subtitle`	Sub-heading
`.card-text`	Paragraph text
`.card-header` / `.card-footer`	Top / bottom sections
`.card-img-top` / `.card-img-bottom`	Images
`.card-img-overlay`	Text placed on top of an image
`.card-group`	Equal-height cards side by side
`.list-group-flush`	Borderless list inside a card

```
<div class="card" style="width: 18rem;">
  <img src="pic.jpg" class="card-img-top" alt="...">
  <div class="card-body">
    <h5 class="card-title">Card Title</h5>
    <p class="card-text">Some quick example text.</p>
    <a href="#" class="btn btn-primary">Read More</a>
  </div>
  <div class="card-footer text-muted">Updated 2 mins ago</div>
</div>
```

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4.13 Modal

A modal is a dialog box that appears on top of the page with a dark backdrop. It blocks interaction with the page until closed.

Class / Attribute	Purpose
`.modal`	Outer wrapper (hidden by default)
`.modal-dialog`	Dialog positioning box
`.modal-content`	The white box
`.modal-header` / `.modal-title`	Header with title
`.modal-body`	Main content
`.modal-footer`	Buttons area
`.fade`	Fade in/out animation
`.modal-dialog-centered`	Vertically centers the dialog
`.modal-sm` / `.modal-lg` / `.modal-xl`	Dialog size
`.modal-dialog-scrollable`	Scrolls the body when content is long
`data-bs-toggle="modal"`	Opens the modal
`data-bs-target="#id"`	Points to the modal's ID
`data-bs-dismiss="modal"`	Closes the modal

```
<!-- Trigger button -->
<button class="btn btn-primary" data-bs-toggle="modal"
        data-bs-target="#myModal">
  Open Modal
</button>

<!-- The modal -->
<div class="modal fade" id="myModal" tabindex="-1">
```

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```
    <div class="modal-dialog modal-dialog-centered">
      <div class="modal-content">

        <div class="modal-header">
          <h5 class="modal-title">Confirm Action</h5>
          <button type="button" class="btn-close"
                  data-bs-dismiss="modal"></button>
        </div>

        <div class="modal-body">
          Are you sure you want to delete this item?
        </div>

        <div class="modal-footer">
          <button class="btn btn-secondary"
                  data-bs-dismiss="modal">Cancel</button>
          <button class="btn btn-danger">Delete</button>
        </div>

      </div>
    </div>
  </div>
```

5. Bootstrap Utilities

Utility classes are small single-purpose classes that do one thing — set a color, add margin, change display, etc. They let you style elements without writing custom CSS.

5.1 Colors

Bootstrap has 8 theme colors: `primary`, `secondary`, `success`, `danger`, `warning`, `info`, `light`, `dark`.

Class pattern	What it sets
`.text-primary`, `.text-danger`...	Text color

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`.text-body`, `.text-muted`, `.text-white`, `.text-black-50`	Special text colors
`.text-opacity-25` … `.text-opacity-100`	Text transparency
`.text-bg-primary`	Background color + matching readable text color
`.link-primary`	Link color

```
<p class="text-primary">Blue text</p>
<p class="text-danger">Red text</p>
<p class="text-muted">Grey secondary text</p>
<span class="text-bg-warning p-2">Warning badge style</span>
```

5.2 Background

Class	Effect
`.bg-primary` … `.bg-dark`	Background color
`.bg-transparent`	No background
`.bg-white` / `.bg-black`	White / black
`.bg-body`, `.bg-body-tertiary`	Theme-aware backgrounds (v5.3+)
`.bg-opacity-10` … `.bg-opacity-100`	Transparency of background
`.bg-gradient`	Adds a subtle linear gradient

```
<div class="bg-primary text-white p-3">Primary background</div>
<div class="bg-success bg-opacity-25 p-3">Faded green background</div>
<div class="bg-dark bg-gradient text-white p-3">Dark gradient</div>
```

5.3 Borders

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Class	Effect
`.border`	Adds a 1px border on all sides
`.border-0`	Removes border
`.border-top`, `.border-end`, `.border-bottom`, `.border-start`	Border on one side only
`.border-primary` … `.border-dark`	Border color
`.border-1` … `.border-5`	Border thickness
`.rounded`, `.rounded-0` … `.rounded-5`	Corner radius
`.rounded-circle`, `.rounded-pill`	Circle / pill shapes
`.rounded-top`, `.rounded-end`	Rounded on one side only

```
<div class="border border-primary border-2 rounded p-3">Box</div>
<div class="border-top border-dark p-2">Top border only</div>
<img src="u.jpg" class="rounded-circle border border-3 border-warning">
```

5.4 Display

These map directly to the CSS `display` property. All of them are responsive: `.d-{breakpoint}-{value}`, e.g. `.d-md-none`.

Class	CSS equivalent
`.d-none`	`display: none`
`.d-block`	`display: block`
`.d-inline`	`display: inline`
`.d-inline-block`	`display: inline-block`

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`.d-flex`	`display: flex`
`.d-inline-flex`	`display: inline-flex`
`.d-grid`	`display: grid`
`.d-table`	`display: table`
`.d-print-none`	Hidden when printing

```
<!-- Hidden on mobile, visible from md upward -->
<div class="d-none d-md-block">Desktop only</div>

<!-- Visible on mobile only -->
<div class="d-block d-md-none">Mobile only</div>

<!-- Flex container -->
<div class="d-flex justify-content-between align-items-center"> ... </div
```

5.5 Overflow

Controls what happens when content is bigger than its container.

Class	CSS equivalent
`.overflow-auto`	Scrollbars only when needed
`.overflow-hidden`	Hides the overflowing content
`.overflow-visible`	Content spills out (default)
`.overflow-scroll`	Always shows scrollbars
`.overflow-x-auto` / `.overflow-y-auto`	Only horizontal / vertical

```
<div class="overflow-auto" style="height: 150px;">
  Long content that will scroll...
</div>
```

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5.6 Position

Class	CSS equivalent
`.position-static`	`position: static` (default)
`.position-relative`	`position: relative`
`.position-absolute`	`position: absolute`
`.position-fixed`	`position: fixed`
`.position-sticky`	`position: sticky`
`.top-0`, `.bottom-0`, `.start-0`, `.end-0`	Offset edges (0–100 + 50)
`.translate-middle`	Centers an absolutely positioned element
`.fixed-top`, `.fixed-bottom`	Pin to top / bottom of viewport
`.sticky-top`	Sticks to the top after scrolling

```
<div class="position-relative" style="height: 200px;">
  <span class="position-absolute top-0 start-0 badge bg-danger">Top-Left<
  <span class="position-absolute bottom-0 end-0 badge bg-primary">Bottom-
</div>
```

5.7 Spacing (Margin and Padding)

The spacing system uses the pattern `{property}{sides}-{size}`.

Part	Values	Meaning
Property	`m`	margin
	`p`	padding
Sides	`t`	top

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	`b`	bottom
	`s`	start (left in LTR)
	`e`	end (right in LTR)
	`x`	left + right
	`y`	top + bottom
	(nothing)	all four sides
Size	`0` to `5`	0, 0.25rem, 0.5rem, 1rem, 1.5rem, 3rem
	`auto`	automatic margin

```
<div class="mt-3">margin-top: 1rem</div>
<div class="px-4">padding left + right: 1.5rem</div>
<div class="p-0">no padding</div>
<div class="mx-auto">centered horizontally</div>
<div class="mb-md-5">margin-bottom 3rem from md upward</div>

<!-- Gap utilities for flex/grid -->
<div class="d-flex gap-3"> ... </div>
```

> **Remember:** `m` = margin (outside space), `p` = padding (inside space). The number goes from 0 (none) to 5 (largest).

5.8 Text Utilities

Class	Effect
`.text-start` / `.text-center` / `.text-end`	Alignment
`.text-wrap` / `.text-nowrap`	Wrapping behaviour
`.text-break`	Breaks long words

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`.text-truncate`	Adds "…" when text overflows
`.text-lowercase` / `.text-uppercase` / `.text-capitalize`	Letter case
`.fw-bold` / `.fw-semibold` / `.fw-light`	Font weight
`.fst-italic` / `.fst-normal`	Font style
`.fs-1` … `.fs-6`	Font size
`.lh-1` / `.lh-sm` / `.lh-lg`	Line height
`.text-decoration-underline` / `-line-through` / `-none`	Decoration
`.font-monospace`	Monospace font

```
<p class="text-center text-uppercase fw-bold">Centered bold heading</p>
<p class="text-truncate" style="max-width: 200px;">A very long line of te
<span class="fs-3 lh-lg">Large text with big line-height</span>
```

5.9 Vertical Align

Used mainly for inline elements, table cells, and flex items.

Class	Effect
`.align-baseline`	Aligns to the text baseline
`.align-top`	Aligns to the top
`.align-middle`	Aligns to the middle
`.align-bottom`	Aligns to the bottom
`.align-text-top` / `.align-text-bottom`	Aligns with the text top / bottom

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`.align-items-start / center / end`	Flex container: align all items vertically
`.align-self-start / center / end`	Flex item: align one item vertically

```
<span class="align-middle">Middle-aligned text</span>

<div class="d-flex align-items-center" style="height: 100px;">
  <span>Vertically centered in the box</span>
</div>
```

Note: `.align-top`, `.align-middle`, and `.align-bottom` only work on *inline*, *inline-block*, *inline-table*, and table-cell elements. For flex layouts, use `.align-items-*` or `.align-self-*` instead.

6. Quick Summary & Best Practices

- **Bootstrap** = HTML + CSS + JS framework for fast, responsive design.
- **Layout** = container → row → columns (12 per row), with breakpoints `sm md lg xl xxl`, plus `offset-*` and `order-*`.
- **Content** = typography, tables (`.table`), images (`.img-fluid`), and forms (`.form-control`).
- **Components** = navbar, navs/tabs, dropdowns, buttons, button groups, breadcrumb, pagination, badges, alerts, progress bars, accordion, cards, modals.
- **Utilities** = single-purpose classes for color, background, border, display, overflow, position, spacing, text, and vertical alignment.

Best Practices

- Always include the viewport meta tag for proper responsive behaviour.
- Use `.container` to wrap rows; never place a `.row` directly in the body.

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- Prefer utility classes over custom CSS for small tweaks.
- Use semantic HTML together with Bootstrap classes (e.g., `<nav>` for navigation).
- Test on real devices or use browser dev tools to check breakpoints.
- Keep accessibility in mind: add `aria-*` attributes where Bootstrap requires them.
- Customize Bootstrap via Sass variables instead of overriding CSS with `!important`.
- Only include the components you need to keep the file size small.

> **Golden rule:** Bootstrap classes are applied left-to-right in the class attribute. If two classes conflict (e.g. `mt-2` and `mt-5`), the one defined later in the Bootstrap CSS file wins — so always check the documentation when combining utilities.

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